Real-World Case: ExxonMobil (XOM)

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I. Executive Summary

We valued ExxonMobil (XOM) using a discounted cash flow (DCF) framework based on normalized mid-cycle operating performance. We based our prediction on realistic mid-cycle margins and reinvestment levels, adjusting for extreme years such as the 2020 COVID collapse and 2022 oil spike due to the company's susceptibility to oil price volatility.

ExxonMobil's estimated enterprise value is approximately $612.4 billion using moderate base-case assumptions that include a 10.78% normalized EBIT margin, 0.59% change in net working capital ratio, 25.87% normalized tax rate, 5.78% weighted cost of capital (WACC), and a 2.00% terminal growth rate. After factoring in net debt from 2024, this calculation implies an intrinsic equity value of roughly $635.5 billion, or approximately $145.98 per share. At the current market price of $150.68 per share as of the February 18 closing price, ExxonMobil's implied enterprise value is slightly above our intrinsic estimate. Sensitivity analysis across reasonable discount rates (5.50% to 6.50%) and terminal growth assumptions (2.00% to 3.00%) produces intrinsic values per share ranging from approximately $124.11 to $199.93.

Overall, under mid-cycle assumptions, ExxonMobil appears to be trading near its fundamental value, with a modest premium relative to our base-case estimate. The current market valuation likely reflects expectations of sustained capital discipline, resilient margins, and a continued strength in global energy demand.

II. Data, key assumptions, and inputs

We constructed a discounted cash flow (DCF) model based on historical financial data drawn from the company's Form 10-K filings from the past ten years (2014-2024) and applied normalized mid-cycle operating assumptions to estimate ExxonMobil's fundamental value. A key component of our analysis was to identify sustainable operating performance rather than rely on peak or trough earnings driven by short-term oil price fluctuations given Exxon's exposure to commodity price volatility. As a result, when calculating the averages to be used in our projections, we excluded the numbers for years 2020 and 2022 due to the COVID-19 pandemic and the oil price spike, respectively (Exhibit B).

From the historical financial data, we were able to calculate the average growth rate, EBIT margin, tax rate, ROIC, CapEx, D&D, and change in NWC (Exhibit A). Using all <u>but the average CapEx as a percentage of revenue</u> values, we were able to project the FCFF for years 2025-2035, keeping the averages normalized through the years (Exhibit C). Note that we did not use the 5.57% average CapEx to project future values because we believe it to be economically inconsistent with our assumption of continued growth. CapEx of 5.57% is less than D&D of 7.37%, suggesting disinvestment and shrinkage of the asset base. However, we assumed a growth rate of 2.08% through the years and a terminal growth rate of 2%, which must be supported by reinvestment. While the 5.57% is not mechanically incorrect, it does not support growing revenue and is representative of the cyclical nature of the oil industry.

To combat the issue of CapEx < D&D, which does not support continued growth, we instead used the growth theory to mathematically estimate CapEx. For this approach, we calculate reinvestment, yearly ROIC, and average ROIC, relying on **Invested Capital = Total Debt + Total Equity – Cash & Cash Equivalents**, **ROIC = Invested Capital/NOPAT**, and **Reinvestment = Growth/ROIC * NOPAT** equations.

From here, we projected future reinvestment and arrived at CapEx for years 2025-2035 with **CapEx = Reinvestment + D&D – Change in NWC** equation. Finally, with all necessary projections, we calculated the FCFF for 2025-2035.

We used the WACC approach to calculate the present values of the FCFF. First, we calculated the market value of equity of $655,910.04 billion, using the market close stock price of $150.68[1] and 4,353 million shares outstanding as of 2024. The shares outstanding was based on the 8,019 million shares issued subtracted by the 3,666 million of common stock held in treasury in 2024. Then, we calculated the total debt of $41,710 million, using the notes and loans payable and long-term debt from the balance sheet. With the CAPM model to calculate the cost of equity, we used a beta of 0.36[1], market risk premium of 5.5%, and risk-free rate of 4.05%[2]. With the 2024 interest expense of $996 million from the 10-K, we calculated the cost of debt. With these inputs, we arrived at the WACC of 5.78% to be used in the discounting (Exhibit D). Finally, we used a terminal growth rate of 2% to discount the terminal value and arrived at an enterprise value of $612,433.41 billion and an intrinsic price of **$145.98** per share (Exhibit E).

III. Intuitive description of the analysis done

Our valuation framework is based on the premise that ExxonMobil's intrinsic value equals the present value of the sustainable free cash flows it is expected to generate over time. Given the company's exposure to commodity price volatility, we avoided extrapolating peak (2022 EBIT of $78.6 billion) or trough (2020 negative EBIT of $27.7 billion) performance shown in Exhibit A. Instead, we normalized operating assumptions to reflect mid-cycle conditions, including an EBIT margin of 10.78%, a D&D-to-revenue ratio of 7.37%, and a normalized tax rate of 25.87%, as summarized in Exhibit B.

Using these normalized inputs, we projected revenues growing at 2.08% annually through 2035, generating EBIT of $37.3 billion in 2025E, rising to $45.9 billion by 2035E (see Exhibit C). Free cash flow to the firm (FCFF) increases from $21.7 billion in 2025E to $26.7 billion in 2035E, reflecting disciplined reinvestment consistent with long-run returns on invested capital. Rather than relying on historical capital expenditures that may reflect cyclical underinvestment, reinvestment was derived from the growth–ROIC relationship to ensure that projected expansion is economically supported.

These projected cash flows were discounted at a weighted average cost of capital (WACC) of 5.78%, calculated using a 4.05% risk-free rate, 5.5% equity risk premium, and 0.36 beta (see Exhibit D). Applying a terminal growth rate of 2.0%, we estimated a terminal value of $720.9 billion, which, when discounted, contributed $411.2 billion to enterprise value. The resulting enterprise value of $612.4 billion leads to an equity value of $635.5 billion and an implied intrinsic price of $145.98 per share, as shown in Exhibit E.

To assess valuation robustness, we conducted sensitivity analysis on both discount rate and terminal growth assumptions. As shown in Exhibit F, varying WACC between 5.5% and 6.5% and terminal growth between 2.0% and 3.0% results in implied share values ranging from $124 to $200, demonstrating the model's sensitivity to long-run discounting assumptions. Similarly, adjusting beta between 0.30 and 0.50 produces an intrinsic value range of $124 to $158 per share, as presented in Exhibit G.

Overall, the analysis reflects the interaction between normalized profitability, disciplined reinvestment, and long-term growth within a cyclical commodity industry. The resulting valuation suggests that ExxonMobil's intrinsic value is driven primarily by sustainable mid-cycle cash flow generation rather than short-term fluctuations in oil prices.

IV. Conclusion

Using our normalized mid-cycle framework, we calculate ExxonMobil's basic enterprise value to be $612.4 billion, which corresponds to an intrinsic equity value of roughly $635.5 billion, or $145.98 per share. At $150.68 per share, ExxonMobil's current market price indicated that it is trading over our base-case intrinsic estimate.

Our valuation reflects sustainable operating performance rather than peak or trough commodity conditions. By normalizing margins, reinvestment, and tax assumptions, and by deriving reinvestment from the growth-ROIC relationship, the model isolates ExxonMobil's long-run cash flow generating capacity under mid-cycle oil price conditions. Since the terminal value component accounts for most of the company value, long-term assumptions about discount rates and perpetual growth and crucial.

Sensitivity analysis shows that even small adjustments to terminal growth or WACC have a significant impact on valuation results. Intrinsic value decreases near the lower end of the modeled range under more conservative assumptions (higher discount rate or lower terminal growth). More optimistic assumptions result in a considerable increase in intrinsic value. This range suggests that expectations of strong global energy demand, stable margins, and ongoing capital discipline, possibly bolstered by structurally tighter supply conditions, are likely reflected in ExxonMobil's present market valuation.

In conclusion, ExxonMobil appears to be trading near its fundamental value under disciplined mid-cycle assumptions. The firm's valuation is therefore highly sensitive to long-term energy price expectations and required return assumptions, rather than short-term earning fluctuations.

V. EXHIBITS

Exhibit A: Historical Financial Summary (2014-2024)

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Revenue	339,247	334,697	398,675	276,692	178,574	255,583	279,332	237,162	200,628	239,854	367,647
EBIT	49,869	53,632	78,551	32,181	(27,725)	20,886	31,719	19,275	8,422	22,277	51,916
NOPAT	35,906	38,079	58,128	24,526	(22,457)	15,456	21,886	20,432	8,843	16,708	33,745
CapEx	24,306	21,919	18,407	12,076	17,282	24,361	19,574	15,402	16,163	26,490	32,952
D&D	23,442	20,461	24,040	20,607	46,009	18,998	18,745	19,893	22,308	18,048	17,297
Change in NWC	(8,676)	(41)	(19,688)	4,162	(1,653)	923	(2,566)	(649)	(182)	(3,113)	(4,932)
FCFF	43,718	33,662	83,449	28,895	7,923	57,892	23,623	25,572	15,170	11,379	23,022

Exhibit A presents ExxonMobil's historical operating and cash flow performance from 2014 through 2024, including revenue, earnings before interest and taxes (EBIT), net operating profit after tax (NOPAT), capital expenditures (CapEx), depreciation and depletion (D&D), change in net working capital (NWC), and free cash flow (FCFF). The data illustrate the company's cyclical earnings pattern, particularly the COVID-19 downtown in 2020 and the oil-price-driven peak in 2022 and provide the basis for the normalized mid-cycle assumptions used in the DCF valuation.

Exhibit B: Normalized Mid-Cycle Assumptions

Mid-Cycle Ratio	Percentage
EBIT margin	10.78%
D&D / Revenue	7.37%
Change in NWC / Revenue	0.59%
Normalized Tax Rate	25.87%

Exhibit B summarizes the normalized mid-cycle operating assumptions derived from historical data, excluding extreme years (2020 and 2022). It presents sustainable estimates for EBIT margin, depletion and depreciation, change in net working capital, and tax rate, which form the foundation of the forward-looking five-year DCF forecast.

Exhibit C: Ten-Year Forecast (2026-2035)

Year	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
Revenue	346,300	353,500	360,849	368,351	376,009	383,826	391,806	399,952	408,267	416,755	425,419
Growth	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%	2.08%
EBIT	37,323	38,099	38,891	39,700	40,525	41,368	42,228	43,106	44,002	44,917	45,851
NOPAT	27,667	28,242	28,829	29,428	30,040	30,665	31,302	31,953	32,617	33,295	33,988
Reinvestment	5,947	6,070	6,197	6,325	6,457	6,591	6,728	6,868	7,011	7,157	7,305
CapEx	29,419	30,030	30,655	31,292	31,943	32,607	33,285	33,977	34,683	35,404	36,140
D&D	25,530	26,061	26,603	27,156	27,721	28,297	28,885	29,486	30,099	30,725	31,363
Δ NWC	2,059	2,101	2,145	2,190	2,235	2,282	2,329	2,377	2,427	2,477	2,529
FCFF	21,720	22,171	22,632	23,103	23,583	24,073	24,574	25,085	25,606	26,139	26,682

Exhibit C presents the projected operating performance and free cash flows for 2026-2035 based on normalized mid-cycle assumptions. Revenue growth, operating margins, reinvestment, and working capital are modeled to reflect ExxonMobil's mature industry position and cyclical risk profile.

Exhibit D: WACC Calculation

	Percentage
Risk-free rate (rf)	4.05%
Equity risk premium (ERP)	5.50%
Beta	0.36
Cost of equity	6.03%
Cost of debt	2.39%
Equity Weight	94.02%
WACC	**5.78%**

Exhibit D presents the derivation of ExxonMobil's weighted average cost of capital using the Capital Asset Pricing Model (CAPM). The cost of equity is estimated based on the risk-free rate, equity risk premium, and beta, while the cost of debt reflects the company's interest expense and capital structure. The resulting WACC assumptions form the discount rate used in the DCF valuation.

Exhibit E: DCF Valuation Summary

Year	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
PV (FCF)	21,720	20,961	20,228	19,522	18,839	18,181	17,546	16,933	16,341	15,770	15,218.7
TV											720,890.8
PV (TV)											411,175.0
EV											612,433.4
Net debt											41,710
Equity value											635,462.4
Share price											145.98

Exhibit E presents the discounted cash flow valuation of ExxonMobil, including the present value of projected free cash flows and the terminal value. Using the normalized operating assumptions and base-case WACC and terminal growth rate, the model produces the estimated fundamental enterprise value of the firm.

Exhibit F: Sensitivity Analysis – WACC and Terminal Growth

WACC / Terminal growth	2.00%	2.50%	3.00%
5.50%	$156.67	$174.70	$199.93
5.78%	$145.81	$160.72	$180.99
6.00%	$138.35	$151.31	$168.59
6.50%	$124.11	$133.77	$146.19

Exhibit F presents the sensitivity of ExxonMobil's enterprise value to changes in the weighted average cost of capital and terminal growth rate. The analysis illustrates how valuation outcomes vary under conservative, moderate, and higher-risk assumptions, highlighting the impact of discount rate and long-run growth on intrinsic value.

Exhibit G: Sensitivity Analysis – Beta

Beta	Intrinsic Price

0.30	$158.15
0.36	$145.98
0.40	$138.93
0.50	$124.13

Exhibit G presents the sensitivity of ExxonMobil's enterprise value to changes in the weighted average cost of capital and terminal growth rate. The analysis illustrates how valuation outcomes vary under conservative, moderate, and higher-risk assumptions, highlighting the impact of discount rate and long-run growth on intrinsic value.